

IILM UNIVERSITY, GREATER NOIDA, U.P.

SCHOOL OF COMPUTER SCIENCE AND ENGINEERING

INTERNSHIP REPORT

ON

**PREDICTION OF AGRICULTURE CROP PRODUCTION
IN INDIA**

Submitted in partial fulfilment of the requirement of degree of

B.Tech CSE (Artificial Intelligence and Machine Learning)

Submitted To

Mrs. Soniya Sharma

School of Computer Science and Engineering

IILM University, Greater Noida, U.P.

Submitted By

YASH ARVIND

Roll Number: 25SCS1003002423

Batch: 2025-29 | 2nd Year, 3rd Semester

Internship Organization: UpSkill Campus / UniConverge Technologies Pvt. Ltd.

Internship Domain: Data Science and Machine Learning

Internship Period: 20-07-2026 to 20-08-2026

CANDIDATE'S DECLARATION

I, Yash Arvind, hereby declare that the internship report titled “Prediction of Agriculture Crop Production in India” is my original work carried out during the internship program in the domain of Data Science and Machine Learning. The work presented in this report was completed as part of the internship requirements and has not been submitted elsewhere for the award of any academic qualification.

The project work involved the preparation of a historical agricultural dataset, development of supervised machine learning features, training and comparison of multiple regression models, and evaluation of their predictive performance. Wherever external datasets, documentation, or learning resources were used, they have been acknowledged appropriately.

Signature: _____

Yash Arvind

Roll No.: 25SCS1003002423

ACKNOWLEDGEMENT

I express my sincere gratitude to UpSkill Campus and UniConverge Technologies Pvt. Ltd. for providing the internship opportunity and practical exposure in Data Science and Machine Learning. The internship provided an opportunity to connect classroom concepts with an applied machine learning project.

I am thankful to the mentors, instructors, and coordinators associated with the internship program for their guidance and learning resources. I also express my gratitude to Mrs. Soniya Sharma and the School of Computer Science and Engineering, IILM University, for their academic guidance and support.

Finally, I am grateful to my family and peers for their encouragement and support throughout the internship period.

Yash Arvind

Roll No.: 25SCS1003002423

INTERNSHIP CERTIFICATE



The certificate confirms successful completion of the internship from 20-07-2026 to 20-08-2026 and identifies the internship domain as Data Science and Machine Learning. It also states that the program was conducted in collaboration with AICTE and UniConverge Technologies & upSkill Campus.

INTERNSHIP OFFER LETTER



C-56/11, Ground floor, Near IT Stellar Park, Sector-62, Noida, U.P. - 201309

Email: support@upskillcampus.com, Ph. No. 9354068856

Internship Offer Letter

Date: **20-07-2026**

Name: **Yash Arvind**

Email: **darcdavelgaming@gmail.com**

Location: **Remote**

Dear **Yash Arvind**,

We are pleased to inform you that **Upskill Campus** alongwith its industry Partner **UniConverge Technologies Pvt. Ltd.** has agreed to provide you internship as **Data Science and Machine Learning Intern**

Your appointment of service is based on the following terms and conditions.

1. The internship shall start from **20-07-2026**
2. **Internship:** The internship period will be 4 to 6 weeks, starting from the date of joining.
3. **Working Hours:** The internship runs from 10:00 AM to 6:00 PM.

On behalf of both the Company, we wish you every success in your position and trust that our relationship will be long and mutually rewarding.

Sincerely

For Upskill Campus and UniConverge Technologies Pvt. Ltd.



Prabha Singh
HR Manager

The offer letter records the role as Data Science and Machine Learning Intern, with the internship starting on 20-07-2026. It specifies a 4–6 week internship period and remote location.

ABSTRACT

Agricultural production is influenced by changing cultivation area, historical yield, weather, resource availability, and other factors. Data-driven forecasting can support planning by estimating future production from historical observations. This project develops a machine learning based approach for predicting agricultural crop production in India using historical crop-level production, area, and yield information.

The dataset contains 55 crop and foodgrain categories with yearly production, area, and yield values for the period 2006–07 to 2010–11. To convert the time-series style data into a supervised learning problem, lag-based features were constructed using the previous year’s production, area, and yield, with the subsequent year’s production used as the target. Linear Regression, Ridge Regression, Random Forest Regression, and Gradient Boosting Regression were trained and compared using Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and R^2 .

Among the evaluated models, Linear Regression achieved the best overall validation performance, with MAE 35.1456, RMSE 57.1770, and R^2 0.9384. Gradient Boosting was a close second in R^2 , while Ridge and Random Forest produced comparatively higher errors on this dataset. The results demonstrate how relatively simple historical indicators can provide a useful baseline for agricultural production forecasting.

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1. INTRODUCTION

1.1 Background

Agriculture is a major application area for data science because production varies across crops and years. Historical production records can be analyzed to identify patterns and construct predictive models. Machine learning provides a flexible way to learn relationships between historical agricultural indicators and subsequent production values.

1.2 Internship Context

The internship was completed in the domain of Data Science and Machine Learning through UpSkill Campus in collaboration with UniConverge Technologies Pvt. Ltd. The certificate records the internship period as 20-07-2026 to 20-08-2026. The offer letter identifies the role as Data Science and Machine Learning Intern.

1.3 Objectives of the Internship

- Gain practical exposure to machine learning workflows.
- Understand how raw datasets are prepared for modelling.
- Apply regression algorithms to an applied prediction problem.
- Compare multiple models using quantitative evaluation metrics.
- Develop the ability to document an end-to-end machine learning project.

1.4 Project Objective

The specific objective of this project is to predict future crop production using historical production, cultivated area, and yield information. The project emphasizes a reproducible machine learning workflow rather than attempting to represent every real-world agricultural variable.

2. ORGANIZATION AND INTERNSHIP OVERVIEW

2.1 UpSkill Campus and UniConverge Technologies

The internship offer letter states that UpSkill Campus, along with its industry partner UniConverge Technologies Pvt. Ltd., provided the internship as a Data Science and Machine Learning Intern. The certificate identifies UpSkill Campus and UniConverge Technologies as part of the program conducted in collaboration with AICTE.

2.2 Internship Period and Role

Item	Details
Candidate	Yash Arvind
Role	Data Science and Machine Learning Intern
Mode / Location	Remote
Start Date	20-07-2026
Completion Date	20-08-2026
Program Duration	Approximately 1 month / 4–6 weeks
Domain	Data Science and Machine Learning

2.3 Work Performed

The project work followed the common machine learning pipeline of understanding the problem, inspecting the dataset, preparing supervised learning features, training regression algorithms, comparing model performance, and documenting the final results. The internship also included learning material and practical project work in the stated domain.

3. PROBLEM STATEMENT

The project addresses the problem of estimating agricultural crop production for a future year from historical crop-level information. Traditional historical records are useful for understanding trends, but a predictive model can provide a systematic estimate from measurable indicators.

3.1 Problem Definition

Given the production, cultivated area, and yield of a crop in a previous year, estimate the production of that crop in the following year. The prediction target is numerical, making this a supervised regression problem.

3.2 Challenges

- The source data is relatively small compared with modern machine learning datasets.
- Only a limited number of yearly observations are available for each crop.
- Agricultural production is affected by variables not included in the dataset, such as weather, irrigation, soil, pests, and market conditions.
- Time-series structure must be preserved when creating training and evaluation examples.

3.3 Scope

The project is a historical-data forecasting prototype. It is intended to demonstrate an end-to-end machine learning approach and model comparison. It should not be interpreted as a production agricultural advisory system because the available data does not include the full range of environmental and operational factors required for such a system.

4. EXISTING AND PROPOSED SOLUTION

4.1 Existing Approach

Agricultural production planning can rely on historical statistics, expert judgement, surveys, and conventional trend analysis. Such approaches remain useful, but they may require manual interpretation and may not provide a consistent computational prediction pipeline.

4.2 Proposed Machine Learning Solution

The proposed solution converts historical crop records into supervised learning samples. For each crop, the previous year's production, area, and yield are used as input features, while the next year's production is treated as the target. Multiple regression models are then trained and evaluated so that the strongest model can be selected based on measured error and goodness of fit.

4.3 Value Addition

- Automates the relationship between historical indicators and production estimates.
- Provides quantitative comparison between different regression algorithms.
- Creates a reproducible preprocessing and modelling pipeline.
- Provides a baseline that can later be extended with weather, soil, irrigation, and regional variables.

4.4 Project Repository

Project code and supporting files were maintained in the GitHub repository:
<https://github.com/Dyza-CZN/Agriculture-Crop-Production-ML---UpskillCampus.git>

5. DATASET AND DATA PREPARATION

5.1 Dataset Description

The project used a historical agriculture dataset containing 55 crop/foodgrain categories. The original table contains five yearly production columns, five yearly area columns, and five yearly yield columns covering 2006–07 through 2010–11, along with a crop category field.

Dataset Property	Value
Number of crop categories	55
Historical years	2006–07 to 2010–11
Original columns	16
Main measures	Production, Area, Yield
Supervised samples after lag construction	220

5.2 Feature Engineering

The original wide-format records were transformed into sequential supervised examples. For each crop and each available transition between consecutive years, the previous year’s production, area, and yield were used as predictors. The following year’s production became the target variable.

Feature	Meaning
Crop	Crop / foodgrain category
Previous_Production	Production in the previous year
Previous_Area	Cultivated area in the previous year
Previous_Yield	Yield in the previous year
Target_Year	Year being predicted
Target_Production	Production in the target year

5.3 Data Quality Considerations

The dataset was inspected and converted into a consistent modelling table. Categorical crop names were encoded for machine learning, while numerical variables were retained as quantitative predictors. Because the available history is short, model results should be interpreted as performance on this dataset rather than as proof of general performance across future agricultural seasons.

6. PROPOSED DESIGN / MODEL

6.1 High-Level Architecture

Stage 1	Stage 2	Stage 3	Stage 4	Stage 5
Historical Agriculture Data	Preprocessing & Feature Engineering	Regression Models	Evaluation & Comparison	Production Prediction

Figure 6.1: High-level workflow of the agriculture crop production prediction system.

6.2 Modelling Approach

Four regression algorithms were implemented and compared: Linear Regression, Ridge Regression, Random Forest Regression, and Gradient Boosting Regression. These models were selected to provide a mix of a simple linear baseline, regularized linear modelling, a tree ensemble, and a boosting-based nonlinear model.

6.3 Evaluation Metrics

Metric	Purpose
MAE	Average absolute prediction error; lower is better.
RMSE	Penalizes larger errors more strongly; lower is better.
R^2	Measures the proportion of target variance explained by the model; higher is better.

6.4 Model Selection

Model selection was based on the combination of MAE, RMSE, and R^2 rather than a single metric. On the recorded evaluation results, Linear Regression produced the lowest MAE and RMSE and the highest R^2 , making it the strongest model among the tested alternatives for this project.

7. IMPLEMENTATION

7.1 Technology Stack

Component	Technology / Tool
Programming Language	Python
Data Processing	Pandas, NumPy
Machine Learning	Scikit-learn
Visualization	Matplotlib
Development Environment	Jupyter Notebook / Python environment
Version Control	Git and GitHub

7.2 Implementation Steps

1. Load the historical agriculture dataset.
2. Clean and standardize crop names and numerical fields.
3. Transform the yearly columns into lag-based supervised samples.
4. Separate input features and production target.
5. Encode the crop category for modelling.
6. Train the four regression models.
7. Evaluate predictions using MAE, RMSE, and R^2 .
8. Compare the models and select the strongest result.
9. Save the prepared data and model results for reproducibility.

7.3 Reproducibility

The project repository contains the machine learning project files, including the prepared data and model comparison outputs. The workflow can therefore be rerun and extended with additional features or algorithms.

8. PERFORMANCE TEST AND RESULTS

8.1 Test Plan / Test Cases

Test Case	Expected Outcome
Dataset loading	All expected crop records and numerical fields are loaded.
Feature construction	Lagged production, area, and yield features are created for consecutive years.
Model training	All four regression models train successfully.
Metric calculation	MAE, RMSE, and R^2 are produced for every model.
Model comparison	A best-performing model can be identified from the recorded metrics.

8.2 Performance Outcome

Model	MAE	RMSE	R^2
Linear Regression	35.1456	57.1770	0.9384
Gradient Boosting	37.8912	57.2286	0.9383
Ridge Regression	38.6284	66.1728	0.9175
Random Forest	43.9643	85.4855	0.8624

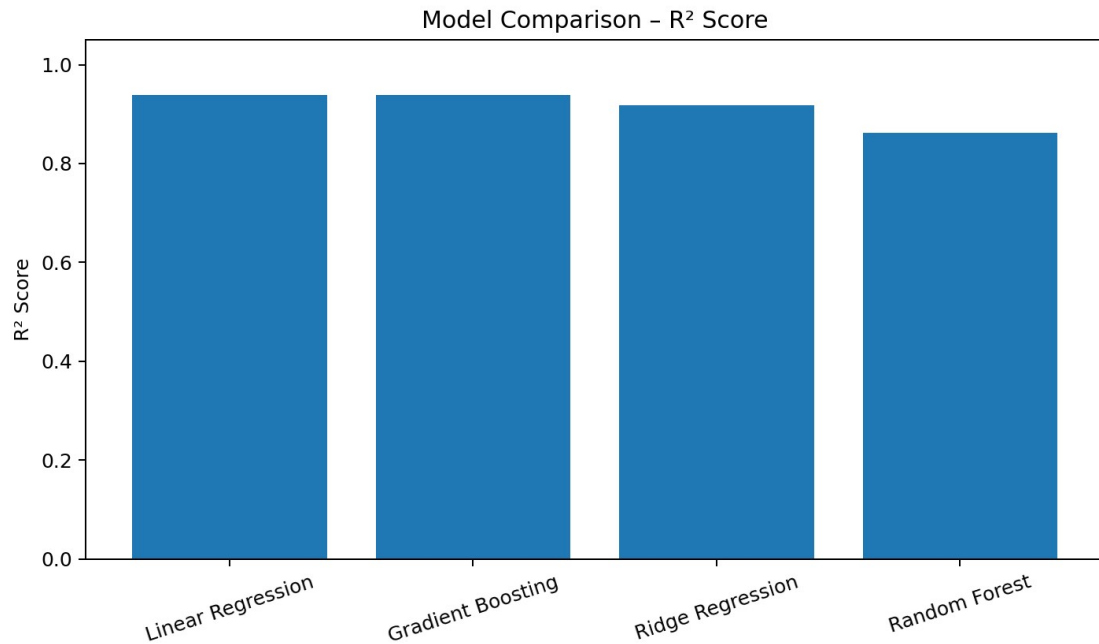


Figure 8.1: R^2 comparison of the evaluated regression models

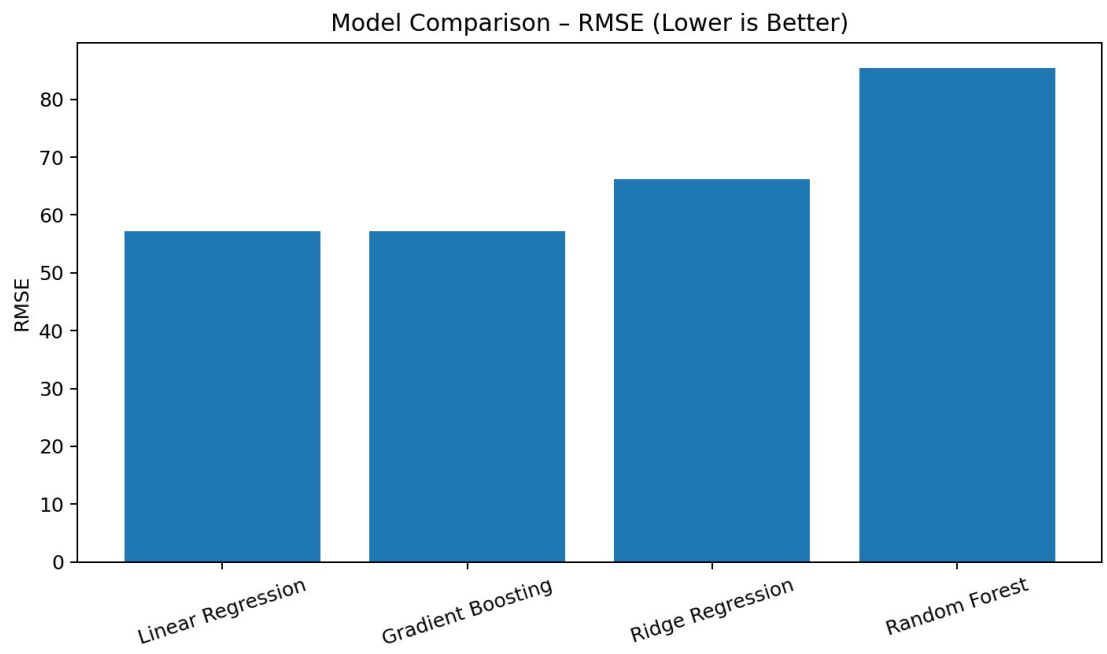


Figure 8.2: RMSE comparison of the evaluated regression models

8.3 Result Interpretation

Linear Regression achieved MAE 35.1456, RMSE 57.1770, and R^2 0.9384. Gradient Boosting produced a very similar R^2 of 0.9383 and RMSE of 57.2286, but its MAE was slightly higher at 37.8912. Ridge Regression achieved R^2 0.9175 with MAE 38.6284 and RMSE 66.1728. Random Forest produced R^2 0.8624, MAE 43.9643, and RMSE 85.4855.

For this dataset, the comparatively strong linear-model performance suggests that the selected historical predictors contain a substantial linear relationship with subsequent production. The result also illustrates why model comparison is important: a more complex ensemble model does not automatically outperform a simpler baseline when the dataset is small and structured in this way.

8.4 Limitations of the Evaluation

The evaluation reflects the supplied historical dataset and the implemented preprocessing pipeline. The dataset contains only five yearly observations per crop before transformation, so the effective amount of temporal information is limited. Broader validation across more years and regions would be required before using such a model for operational agricultural forecasting.

9. MY LEARNINGS

- Understanding the complete machine learning lifecycle from problem definition to evaluation.
- Practical use of Pandas for data inspection and transformation.
- Creating lag-based features for a time-dependent prediction problem.
- Training and comparing multiple regression algorithms using Scikit-learn.
- Interpreting MAE, RMSE, and R^2 instead of relying only on a single accuracy value.
- Understanding that dataset size and feature coverage strongly affect model reliability.
- Using GitHub to organize and present project work.

9.1 Career Relevance

The project strengthened practical skills in Python, data preparation, supervised learning, model evaluation, and technical documentation. These skills form a foundation for further work in machine learning, data analytics, forecasting, and applied artificial intelligence.

10. FUTURE WORK SCOPE

- Add more years of historical data to improve temporal coverage.
- Include state, district, soil, rainfall, temperature, irrigation, and fertilizer-related features.
- Experiment with time-series models and sequence-based neural networks when sufficient data is available.
- Perform stronger chronological cross-validation across multiple historical periods.
- Develop crop-specific and region-specific models.
- Create a simple dashboard or API for interactive production forecasts.
- Add uncertainty estimates so that predictions are presented as ranges rather than single values.

10.1 Recommended Extension

The most valuable next step would be to combine production history with weather and regional agricultural variables. This would reduce dependence on production history alone and make the model more representative of the factors that influence real-world crop output.

11. CONCLUSION

This internship project demonstrated an end-to-end machine learning workflow for predicting agricultural crop production in India from historical production, area, and yield information. The dataset was transformed into supervised learning samples using lagged features, and four regression models were trained and compared.

Linear Regression was the best-performing model among the tested approaches on the recorded evaluation results, achieving an R^2 of 0.9384, MAE of 35.1456, and RMSE of 57.1770. The project also showed that model complexity should be matched to the available data and problem structure. With a larger and richer dataset, the approach can be extended into a more comprehensive agricultural forecasting system.

12. REFERENCES

- [1] UpSkill Campus / UniConverge Technologies Pvt. Ltd., Internship Certificate of Completion, 2026.
- [2] UpSkill Campus / UniConverge Technologies Pvt. Ltd., Internship Offer Letter, 20-07-2026.
- [3] Vijay K. Rohatgi, An Introduction to Probability and Statistics, 3rd Edition.
- [4] Scikit-learn documentation, machine learning algorithms and model evaluation concepts.
- [5] Pandas documentation, data manipulation and analysis library.
- [6] NumPy documentation, numerical computing in Python.
- [7] Matplotlib documentation, plotting and visualization in Python.
- [8] Government of India / data.gov.in agricultural datasets used as the project data source context.

APPENDIX A – SUPPORTING DOCUMENTS

The internship certificate and offer letter supplied for this internship are included in the preceding supporting-document pages. The final academic GitHub submission is intended to contain the internship report, internship presentation, and internship certificate as required by the university evaluation instructions.

APPENDIX B – PROJECT REPOSITORY

Agriculture project GitHub repository: <https://github.com/Dyza-CZN/Agriculture-Crop-Production-ML---UpskillCampus.git>